



IBM Quantum

Single-Qubit Gates & Superposition

Fatima Bint-e-Munir



Qiskit Fall Fest 2026 at University of Engineering & Technology Lahore, Normal Campus

Speaker Introduction

- ABOUT:**
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FALL FEST ROADMAP

You Are Here: Talk 2 of the Series



Intro to QC + Qiskit setup

Why quantum, environment ready



Vectors, matrices, inner products

The math language of qubits



Statevector & Operator

Representing & transforming states in Qiskit



Single-qubit gates + superposition

This talk — X, Y, Z, H, S, T

WE ARE HERE

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60-SECOND RECAP

What You Already Know

From Talk 1 — we'll build directly on these four ideas today.

A qubit is a vector

$|\psi\rangle = c_0|0\rangle + c_1|1\rangle$, with $|c_0|^2 + |c_1|^2 = 1$

Statevector

Qiskit's quantum info Statevec for holds this vector & lets us inspect it

Operator

A matrix that transforms one statevector into another

The Bloch sphere

A picture of every possible single-qubit state as a point on a sphere

From last time

```
from qiskit.quantum_info import Statevector, Operator
sv = Statevector([1, 0]) # the state |0>
print(sv) # Statevector([1.+0.j, 0.+0.j]), dim=(2,1)
```

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Where We're Headed

01	What is a gate?	Matrices, unitarity, rotations
02	The Pauli gates	X, Y, Z — the basic flips
03	Hadamard & superposition	Creating and measuring superposition
04	The phase family	S and T — fine phase turns
05	Putting it together	Building & reading real circuits

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BY THE END OF THIS TALK

Learning Objectives

- Explain what a quantum gate is, mathematically and geometrically
- Write down the matrix for X, Y, Z, H, S, and T — and know what each one does
- Read and write the Qiskit code that applies any of these gates
- Read a single-qubit circuit diagram and predict the resulting state
- Explain superposition correctly — and describe how measurement statistics reveal it

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PART 1

What Exactly Is a Quantum Gate?

Before meeting X, Y, Z, H, S and T individually — what are we even doing when we apply a gate?

FROM SWITCHES TO SPINS

A Familiar Idea, A New Rulebook

CLASSICAL GATE	QUANTUM GATE
NOT	X
$0 \rightarrow 1$ $1 \rightarrow 0$	$X 0\rangle = 1\rangle$ $X 1\rangle = 0\rangle$ $X +\rangle = +\rangle$
- Acts on a definite bit - Output is always 0 or 1 - No notion of "in-between"	- Acts on a vector — including combinations of 0 and 1 - Always reversible (you can always undo it) - Output can be any point on the Bloch sphere

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THE FORMAL DEFINITION

A Gate Is a Unitary Matrix

Remember multiplying a matrix by a vector. A quantum gate is exactly that — with one extra rule.

A gate U transforms a statevector:

$$| \psi' \rangle = U | \psi \rangle$$

Same matrix-vector multiplication as Operator() from last talk — and applied to a quantum state.

The extra rule: **unitarity**

$$U^\dagger U = I$$

$U^\dagger$ is U 's conjugate transpose. This guarantees every gate is reversible and preserves total probability (vector length 1).

Why it matters

Because gates are reversible, quantum computation never "loses doors." Every gate we will see today has an inverse — often itself! This is fundamentally different from classical gates like AND, which throw information away.

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THE GEOMETRIC PICTURE

Gates Are Rotations of the Bloch Vector

Because length is preserved, a gate can only rotate a state around the sphere.

Every state is a point
on the surface of the sphere (a unit-length vector).

Every gate is an angle from some axis
a rotation the state is carried through.

Applying more gates = adding rotations
we'll build circuits by composing simple turns.

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OUR METHOD FOR EVERY GATE

The Recipe We'll Repeat Six Times

Every gate — X, Y, Z, H, S, T — is introduced in four steps. Once you know the pattern, you can decode any new gate yourself.

1

Intro & Matrix

Name, symbol, and the 2x2 matrix that defines it.

2

Bloch View

Which axis it rotates around, and by how much.

3

Qiskit Code

The one-line call, and the circuit diagram it draws.

4

Try It

A worked example — what happens to a real state.

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PART 2

The Pauli Gates

X, Y, Z — the three fundamental flips of a qubit, named after physicist Wolfgang Pauli

X

Y

Z

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X

PAULI GATE

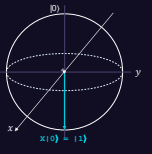
The X Gate — Bit Flip

Flips $|0\rangle \rightleftharpoons |1\rangle$. A π (180°) rotation around the X-axis of the Bloch sphere.

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Nickname

The "quantum NOT gate" — the direct analogue of the classical bit flip, extended to work on superpositions too.



$|0\rangle$

$|1\rangle$

$X|0\rangle = |1\rangle$

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X

PAULI GATE

X in Qiskit

One call adds the gate to the circuit; `qc.draw()` shows it on the wire.

```
Qiskit
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector

qc = QuantumCircuit(1)
qc.x(0) # apply X to qubit 0
qc.draw(output="mpl")
```

Circuit diagram



$q: |0\rangle$ — **X** —

Gate methods take the qubit index, `qc.x(0)` means "apply X to qubit 0." The same pattern — `qc.gate-(qubit)` — works for every single-qubit gate we'll see today.

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X

PAULI GATE

Try It — X on Both Basis States

Matrix multiplication by hand, then the same result straight from Qiskit.

$X|0\rangle = |1\rangle$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The qubit certainly starting at 0 becomes certainly 1.

$X|1\rangle = |0\rangle$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

X undoes itself: applying X twice returns the original state ($X^2 = I$).

Verify with Statevector

```
qc = QuantumCircuit(1)
qc.x(0)
print(Statevector(qc))
# Statevector([0.+0.j, 1.+0.j]), dim=2(2)) — this is |1>
```

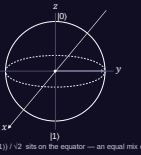
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X

PAULI GATE

Predict Before You Peek



$|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ sits on the equator — an equal mix of 0 and 1.

? We know X flips $|0\rangle$ and $|1\rangle$. So what do you think X does to $|\psi\rangle$, the equal-superposition state? Does it change at all?

Think about it — we'll build the intuition together across the next sections.

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Z

PAULI GATE

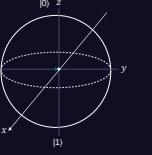
The Z Gate — Phase Flip

Leaves $|0\rangle$ and $|1\rangle$ alone, but flips the sign of $|1\rangle$. A π rotation around the Z-axis.

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Key insight

Z does nothing visible to $|0\rangle$ or $|1\rangle$ alone — its effect only shows up once a qubit is in superposition.



$|0\rangle$

$|1\rangle$

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Z

PAULI GATE

Z in Qiskit

```
Qiskit
qc = QuantumCircuit(1)
qc.h(0) # first search |psi>
qc.z(0) # apply Z to qubit 0
```

Circuit diagram



$q: |0\rangle$ — **H** — **Z** —

Try it: $Z|\psi\rangle = -|\psi\rangle$

```
[1 0] [0.707] = [0.707]
0 -1 [0.707] = [-0.707]
```

`print(Statevector(qc))` — `Statevector([0.707+0.j, -0.707+0.j], dim=2(2))`

This minus sign is a relative phase — invisible in measurement probabilities, but it changes how this qubit will interfere with others later.

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Z PAULI GATE

A Gate You Can't See... Yet

$Z|0\rangle = |0\rangle$ $Z|1\rangle = -|1\rangle$

A global minus sign on a basis state is unobservable — measuring $|1\rangle$ or $-\lvert 1\rangle$ gives "1" with probability 1 either way.
So if a measurement can never tell $|1\rangle$ from $-\lvert 1\rangle$ apart... has Z actually done anything at all?

? Given what you just saw with Z on $|1\rangle$ — what's different about applying Z to a superposition versus to a basis state?

Hint: relative phase between two amplitudes is different from an overall sign on one state.

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Y PAULI GATE

The Y Gate — Bit-Flip + Phase-Flip

Combines what X and Z do. A π rotation around the Y-axis (out of the X-Z plane we've been drawing).

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$Y|0\rangle = i|1\rangle$ $Y|1\rangle = -i|0\rangle$

Notice the bit flips ($0 \leftrightarrow 1$) exactly like X, and a factor of i appears — a phase, like Z. Up to that phase, $Y \approx X$ then Z.

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Y PAULI GATE • STEP 2 OF 2

Y in Qiskit

```

qiskit
qc = QuantumCircuit(1)
qc.y(0) # Apply Y to qubit 0
print(Statevector(qc))
# Statevector([0.+0.j, 0.+1.j]), dims=(2,))

```

Circuit diagram

The X, Y, Z family — one relation to remember

$X = iY Z$ $Y = -iX Z$ $Z = iX Y$

Applying Z then X , then correcting an overall phase i , reproduces Y .

We return to gate composition later.

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SECTION RECAP

Three Axes, Three Gates

- X** Rotate π around X — bit-flip
- Y** Rotate π around Y — bit + phase flip
- Z** Rotate π around Z — phase-flip

All three are their own inverse: $X^2 = Y^2 = Z^2 = I$.

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QUICK BREATH

Where Do "X, Y, Z" Actually Come From?

These matrices are named after Wolfgang Pauli, who introduced them in 1927 — decades before quantum computers — to describe electron spin. An electron's spin can be measured along any axis, and only ever gives two outcomes: "up" or "down." Sound familiar? A qubit's $|0\rangle$ and $|1\rangle$ are the direct descendants of that idea.

Search, take a sip of water, and roll your shoulders — Part 3 (Hadamard & superposition) is where things get genuinely weird.

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PART 3

The Hadamard Gate & Superposition

The gate that makes quantum computing quantum

H

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THE CORE IDEA

What Is Superposition, Really?

A qubit in superposition is still described by a single vector with two nonzero amplitudes at once — not two separate realities.

A useful (imperfect) analogy

A spinning coin, mid-air, is neither heads nor tails — it's a genuine physical mix of both, described by one set of numbers.

Only when it lands — analogous to measurement — does it commit to one outcome.

Whence the analogy breaks: a coin's outcome is just hidden. A qubit isn't decided until it's measured.

The math

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

α and β are complex amplitudes. Both can be nonzero simultaneously — one vector, one state, two amplitudes.

$$|\alpha|^2 + |\beta|^2 = 1$$

The normalization constraint — total probability always sums to 1, no matter how the amplitude is split.

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CLEARING UP A COMMON MIX-UP

"Both at Once" Is a Trap

✗ MISLEADING

"The qubit is secretly 0 AND secretly 1 at the same time, we just don't know which."

This describes classical uncertainty, like a hidden coin — and it makes the wrong experimental predictions. If this were true, interference effects (which quantum algorithms may only simply wouldn't happen).

✓ MORE ACCURATE

"The qubit's state genuinely is the combination $\alpha|0\rangle + \beta|1\rangle$ — with no definite value for 0 or 1 until it's measured."

The amplitudes α, β are real and physical, and can interfere — add up or cancel out — the way waves do. That interference is the resource quantum algorithms exploit.

We'll build the exact amplitude — probability rule in a few slides.

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HADAMARD GATE · STEP 1 OF 3

The Hadamard Gate

Rotates $|0\rangle$ to the equator — an equal superposition. The superposition makes

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

A diagonal rotation

H transforms $|0\rangle$ to halfway between $|0\rangle$ and $|1\rangle$.

$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

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HADAMARD GATE · STEP 1 OF 3

The Hadamard Gate

Rotates $|0\rangle$ to the equator — an equal superposition. The superposition makes

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

A diagonal rotation

H rotates π around the axis exactly halfway between X and Z — that's why it turns a pole into an equator point.

$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

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HADAMARD GATE · STEP 2 OF 3

H in Qiskit

Qiskit

```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector

qc = QuantumCircuit(1)
qc.h(0) # apply H to qubit 0
qc.draw(output="mpl")
```

Circuit diagram

Notice: same pattern as every gate so far — qc.h(0). One consistent API, whatever the gate does underneath.

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HADAMARD GATE · STEP 3 OF 3

Try It — Building $|+\rangle$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{(|0\rangle + |1\rangle)}{\sqrt{2}} = |+\rangle$$

Verify with Statevector

```
qc = QuantumCircuit(1)
qc.h(0)
print(Statevector(qc))
# Statevector([0.70710678+0.j, 0.70710678+0.j], dims=(2,))
# Same magnitude on both amplitudes — equal superposition
```

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► PREDICT BEFORE YOU PEEK

What About $H|1\rangle$?



? H sends $|0\rangle$ to the equator. Where do you think H sends $|1\rangle$ — the same point, or somewhere else on the equator?

Think geometrically: H is a rotation around a fixed diagonal axis.

We just found $H|0\rangle = |x\rangle$ (the equator, right side).

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► FROM AMPLITUDE TO PROBABILITY

The Born Rule

Reveal for the last slide: $H|1\rangle = |y\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ — same equator point family, opposite relative phase. Now, how do amplitudes turn into what we actually observe?

P(outcome) = |amplitude|²

The probability of measuring a given outcome is the squared magnitude of its amplitude.

For $|x\rangle = 1/\sqrt{2}(|0\rangle + |1\rangle)$:

$$P(0) = |1/\sqrt{2}|^2 = 1/2$$

$$P(1) = |1/\sqrt{2}|^2 = 1/2$$

A perfect (quantum) coin flip — 50/50, every time we measure.

Note the amplitude is lost

Squaring throws away sign and phase — that's why $Z|x\rangle = |x\rangle$ measures identically to $|x\rangle$, even though the states are different. The phase only matters again, once more gates act on it.

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► FROM THEORY TO DATA

Let's Actually Measure It

```
from qiskit.primitives import StatevectorSampler
qc = QuantumCircuit(1)
qc.h(0)
qc.measure_all()
sampler = StatevectorSampler()
job = sampler.run(qc, shots=1000)
counts = job.result().get_counts()
```

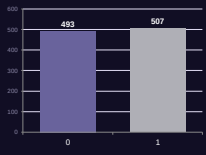
? We'll run this 1,000 times. Given $P(0)=P(1)=1/2$, what split do you expect?

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► THE REVEAL

plot_histogram(counts)

1,000 shots of $H|0\rangle$, measured and binned — statistical noise means it's never exactly 500/500.



Right in line with theory

493/507 is well within the statistical fluctuation expected for 1,000 coin-flip-like trials at $p=0.5$.

This is the experimental signature of superposition: a single qubit, prepared identically every time, genuinely produces both outcomes across many runs.

Plot it yourself — the exact numbers will differ each time you execute this cell.

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► ONE MORE PROPERTY

What Happens If We Apply H Twice?

$q: |0\rangle \xrightarrow{H} \xrightarrow{H}$

? Apply H twice in a row — where does the state end up?

Reveal

```
qc = QuantumCircuit(1)
qc.h(0)
qc.h(0)
print(Statevector(qc))
# Statevector([1.+0.j, 0.+0.j]) → back to |0>, P=1
```

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► LOOKING AHEAD

Superposition Scales Up Fast

Apply H to every qubit in an n -qubit register, and you get an equal superposition of all 2^n possible bitstrings — from a single line of code.

$q0: |0\rangle \xrightarrow{H}$
 $q1: |0\rangle \xrightarrow{H}$
 $q2: |0\rangle \xrightarrow{H}$

3 qubits — 8 states at once, each with equal amplitude. This exponential scaling — n qubits, 2^n amplitudes — is the raw resource that quantum algorithms (a future talk) learn to exploit.

Quick — 3 qubits

```
qc = QuantumCircuit(3)
qc.h([0, 1, 2])
print(Statevector(qc))
# equal superposition of all
# 8 states: (000) ... (111)
```

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PART 4

The Phase Family: S and T

Fine-grained phase turns — the fractional gears behind Z

S **T**

Qiskit logo

ONE GATE TO RULE THEM ALL

The Phase Gate $P(\phi)$

Z, S, and T are all the same gate, just dialed to a different angle.

$$P(\phi) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix}$$

Leaves $|0\rangle$ untouched, and multiplies the amplitude of $|1\rangle$ by $e^{i\phi}$ — a pure rotation of its phase by angle ϕ .

Z $P(\pi) = Z$ (half turn)

S $P(\pi/2) = S$ (quarter turn)

T $P(\pi/4) = T$ (eighth turn)

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S **PHASE GATE · STEP 1 OF 2**

The S Gate — Quarter Turn ($\sqrt{Z}$)

Rotates the $|1\rangle$ phase by 90° . Apply it twice and you get Z exactly.

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

S · **S** = **Z**

Two quarter turns make a half turn — S is literally the square root of Z.

Qiskit logo

S **PHASE GATE · STEP 2 OF 2**

S in Qiskit

```

Qiskit
qc = QuantumCircuit(1)
qc.h(0) # reach |+> first
qc.s(0) # apply S to qubit 0

```

Circuit diagram

$q: |0\rangle \rightarrow H \rightarrow S$

Try it: $S|+\rangle = (|0\rangle + i|1\rangle) / \sqrt{2}$

```

print(Statevector(qc) - Statevector([0.707+0.j, 0.+0.707j]), dim=2,))

```

The $|1\rangle$ amplitude picked up a factor of i — same magnitude (0.707), rotated phase. Measurement probabilities are still exactly 50/50.

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T **PHASE GATE · STEP 1 OF 2**

The T Gate — Eighth Turn ($\sqrt[4]{S}$)

Rotates the $|1\rangle$ phase by 45° — nicknamed the "tribe gate." The finest phase step we'll use.

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

T · **T** = **S** **T**⁴ = **Z**

Four eighth turns make a half turn. T is the square root of S.

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T **PHASE GATE · STEP 2 OF 2**

T in Qiskit

```

Qiskit
qc = QuantumCircuit(1)
qc.h(0) # reach |+> first
qc.t(0) # apply T to qubit 0

```

Circuit diagram

$q: |0\rangle \rightarrow H \rightarrow T$

Try it: $T|+\rangle = (|0\rangle + e^{i\pi/4}|1\rangle) / \sqrt{2}$

```

print(Statevector(qc) - Statevector([0.707+0.j, 0.5+0.5j]), dim=2,))

```

$e^{i\pi/4} = 0.707 + 0.707i$, so the $|1\rangle$ amplitude $0.707e^{i\pi/4}$ works out to 0.5 + 0.5i. Same tick as S, finer angle.

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KEEP THIS HANDY

The Single-Qubit Gate Cheat Sheet



Gate	Matrix	Qiskit	Block action	One-liner
X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	<code>qc.x(q)</code>	π around X	bit flip
Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	<code>qc.y(q)</code>	π around Y	bit + phase flip
Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	<code>qc.z(q)</code>	π around Z	phase flip
H	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	<code>qc.h(q)</code>	π , diagonal axis	makes superposition
S	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	<code>qc.s(q)</code>	$\pi/2$ around Z	quarter phase turn
T	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i(2\pi/8)} \end{bmatrix}$	<code>qc.t(q)</code>	$\pi/4$ around Z	eighth phase turn

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RECAP

Let's Check the List



- ✓ Explain what a quantum gate is, mathematically and geometrically
- ✓ Write down the matrix for X, Y, Z, H, S, and T — and know what each one does
- ✓ Read and write the Qiskit code that applies any of these gates
- ✓ Read a single-qubit circuit diagram and predict the resulting state
- ✓ Explain superposition correctly — and describe how measurement statistics reveal it

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KEEP EXPLORING

Resources



IBM Quantum Learning

Find interactive courses — the source for today's explanations and code

quantum.cloud.ibm.com/learning

Qiskit API documentation

The full reference for every gate, primitive, and visualization function

quantum.cloud.ibm.com/docs

Qiskit source & examples

Review real circuits and contribute to the open-source project

[github.com/qiskit/qiskit](https://github.com/Qiskit/qiskit)

IBM Quantum Composer

Drag-and-drop circuit building in your browser, no install needed

quantum.cloud.ibm.com/composer

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THANK YOU

Questions?



No pressure, no wrong questions — if a gate still feels fuzzy, now's the time.

X

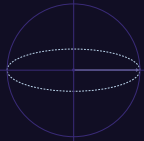
Y

Z

H

S

T



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